

PRESBURGER LOGIC AND SEMI-LINEAR SETS

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2.1

What is Presburger Arithmetic?

Presburger Arithmetic

Presburger arithmetic is the first-order theory of the natural numbers with addition, named after Mojżesz Presburger, who introduced it in 1929.

- The language includes addition and equality but omits multiplication.
- It is strictly weaker than Peano arithmetic, which includes both addition and multiplication.

2.2

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- The language includes addition and equality but omits multiplication.
- It is strictly weaker than Peano arithmetic, which includes both addition and multiplication.
- All formulas in the theory are decidable.
- Presburger Arithmetic is decidable while Peano Arithmetic is not.

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Axioms of Presburger Arithmetic

Universal Axioms

The axioms of Presburger arithmetic are the universal closures of the following formulas:

1. $\neg(0 = x + 1)$
2. $x + 1 = y + 1 \rightarrow x = y$
3. $x + 0 = x$
4. $x + (y + 1) = (x + y) + 1$
5. **Induction Schema:** Let $P(x)$ be any formula in the language of Presburger arithmetic (with x free). Then:

$$(P(0) \wedge \forall x (P(x) \rightarrow P(x + 1))) \rightarrow \forall y P(y)$$

4.1

First-Order Logic (FOL)

What is First-Order Logic?

First-order logic (FOL), also called predicate logic, extends propositional logic by allowing:

- Quantified variables: $\forall x, \exists y$, etc.
- Predicates for properties and relations: e.g., $\text{Man}(x)$, $\text{Mortal}(x)$

4.2

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- Quantified variables: $\forall x, \exists y$, etc.
- Predicates for properties and relations: e.g., $\text{Man}(x)$, $\text{Mortal}(x)$
- Logical sentences like:

$$\forall x(\text{Man}(x) \rightarrow \text{Mortal}(x))$$

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FOL vs Second-Order Logic

Quantification in First-Order Logic

In FOL, we can quantify over individuals, but **not over predicates**:

$\exists x \text{Cube}(x)$ ✓ Valid in FOL

$\exists P P(b)$ Not valid in FOL (only in SOL)

In Second-Order Logic (SOL), predicates can be quantified:

$\exists P \forall x (Px \leftrightarrow (\text{Cube}(x) \vee \text{Tet}(x)))$

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Presburger Logic

First-Order Logic over $(\mathbb{N}, 0, 1, +, =, <)$

- Interpreted over $\mathbb{N} = \{0, 1, 2, 3, \dots\}$
- Allowed operations: addition (+), order (<), and logical connectives:

$$\neg, \wedge, \vee, \forall, \exists$$

- You can express:
 - $\forall x \forall y ((x < y) \Rightarrow \exists z (x < z < y))$
 - “Every number is odd or even”:

$$\forall x \exists y (x = 2y \vee x = 2y + 1)$$

7.1

Presburger Logic Limitations

What You **Cannot** Express

- Multiplication of variables (non-linear terms), e.g.:

$$x \cdot y, \quad x^2$$

- Properties like:
 - Primality: $\text{prime}(x)$
 - Divisibility: $y \mid x$

7.2

Presburger Logic Limitations

What You **Cannot** Express

- Multiplication of variables (non-linear terms), e.g.:

$$x \cdot y, \quad x^2$$

- Properties like:
 - Primality: $\text{prime}(x)$
 - Divisibility: $y \mid x$
- These require **first-order arithmetic with multiplication**.

8 Presburger Logic: Syntax and Semantics

Terms and Formulas

Terms t are built using:

$$t ::= 0 \mid 1 \mid x \mid y \mid t + t$$

Atomic formulas f :

$$f ::= t = t \mid t < t$$

General formulas φ :

$$\varphi ::= f \mid \neg \varphi \mid \varphi \vee \varphi \mid \varphi \wedge \varphi \mid \exists x \varphi \mid \forall x \varphi$$

Semantics: $L(\varphi)$ is the set of all interpretations (assignments to variables) that make φ true.

9.1

What is Decidability?

Formal Definition

A set $S \subseteq \Sigma^*$ is **decidable** if there exists a Turing machine (or equivalently, an algorithm) that halts on every input and:

- Accepts inputs $x \in S$
- Rejects inputs $x \notin S$

9.2

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Formal Definition

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- Rejects inputs $x \notin S$

Intuitive Idea

There is a finite procedure that always terminates and determines whether a given input belongs to the set or not.

10.1

Peano Arithmetic is Undecidable

A Natural Decision Problem

Given a polynomial equation with integer coefficients:

$$p(x_1, x_2, \dots, x_n) = 0$$

Does it have a solution in $\mathbb{N}$?

◇ What is Presburger?
Presburger $\leftrightarrow$ *Semilinear*

> **Why Presburger?**
> CFL >

> Automata based Construction
> Semilinear Sets > Decidability

◇

10.2

Peano Arithmetic is Undecidable

A Natural Decision Problem

Given a polynomial equation with integer coefficients:

$$p(x_1, x_2, \dots, x_n) = 0$$

Does it have a solution in $\mathbb{N}$?

■ $x^2 + y^2 - 1 = 0$

Yes

10.3

Peano Arithmetic is Undecidable

A Natural Decision Problem

Given a polynomial equation with integer coefficients:

$$p(x_1, x_2, \dots, x_n) = 0$$

Does it have a solution in $\mathbb{N}$?

- $x^2 + y^2 - 1 = 0$
- $x^2 + y^2 - 3 = 0$

Yes

No

10.4

Peano Arithmetic is Undecidable

A Natural Decision Problem

Given a polynomial equation with integer coefficients:

$$p(x_1, x_2, \dots, x_n) = 0$$

Does it have a solution in $\mathbb{N}$?

- $x^2 + y^2 - 1 = 0$
- $x^2 + y^2 - 3 = 0$
- $(x + 2)^2 - 61y^2 = 1$

Yes

No

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Smallest Solution Example

Smallest Solution

Consider the equation:

$$(x + 2)^2 - 61y^2 = 1$$

The smallest solution in $\mathbb{N}$ is:

$$x = 1766319047, \quad y = 226123980$$

12.1

Hilbert's Tenth Problem

The Problem

a polynomial equation with integer coefficients:

$$p(x_1, x_2, \dots, x_n) = 0$$

Is there an algorithm that can decide whether it has a solution in $\mathbb{N}$ or $\mathbb{Z}$?

◇ What is Presburger?
Presburger $\leftrightarrow$ Semilinear

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12.2

Hilbert's Tenth Problem

The Problem

a polynomial equation with integer coefficients:

$$p(x_1, x_2, \dots, x_n) = 0$$

Is there an algorithm that can decide whether it has a solution in $\mathbb{N}$ or $\mathbb{Z}$?

Answer (Matiyasevich, 1970)

No. There is no such algorithm.

The problem was solved **in the negative**.

Therefore, **Peano Arithmetic is undecidable**.

◇ What is Presburger?
Presburger $\leftrightarrow$ Semilinear

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◇

13.1

Decision Problem for Presburger Arithmetic

Key Question

Does there exist a finite procedure (algorithm) that, given a statement in Presburger arithmetic, always decides whether it is true or false?

◇ What is Presburger?
Presburger $\leftrightarrow$ *Semilinear*

> **Why Presburger?**
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◇

13.2

Decision Problem for Presburger Arithmetic

Key Question

Does there exist a finite procedure (algorithm) that, given a statement in Presburger arithmetic, always decides whether it is true or false?

Answer

- **Yes!** Presburger arithmetic is decidable.

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Automata-Based Construction

Encoding Variable

Interpret each assignment as a binary string:

$$x \mapsto 00101 \quad y \mapsto 00010$$

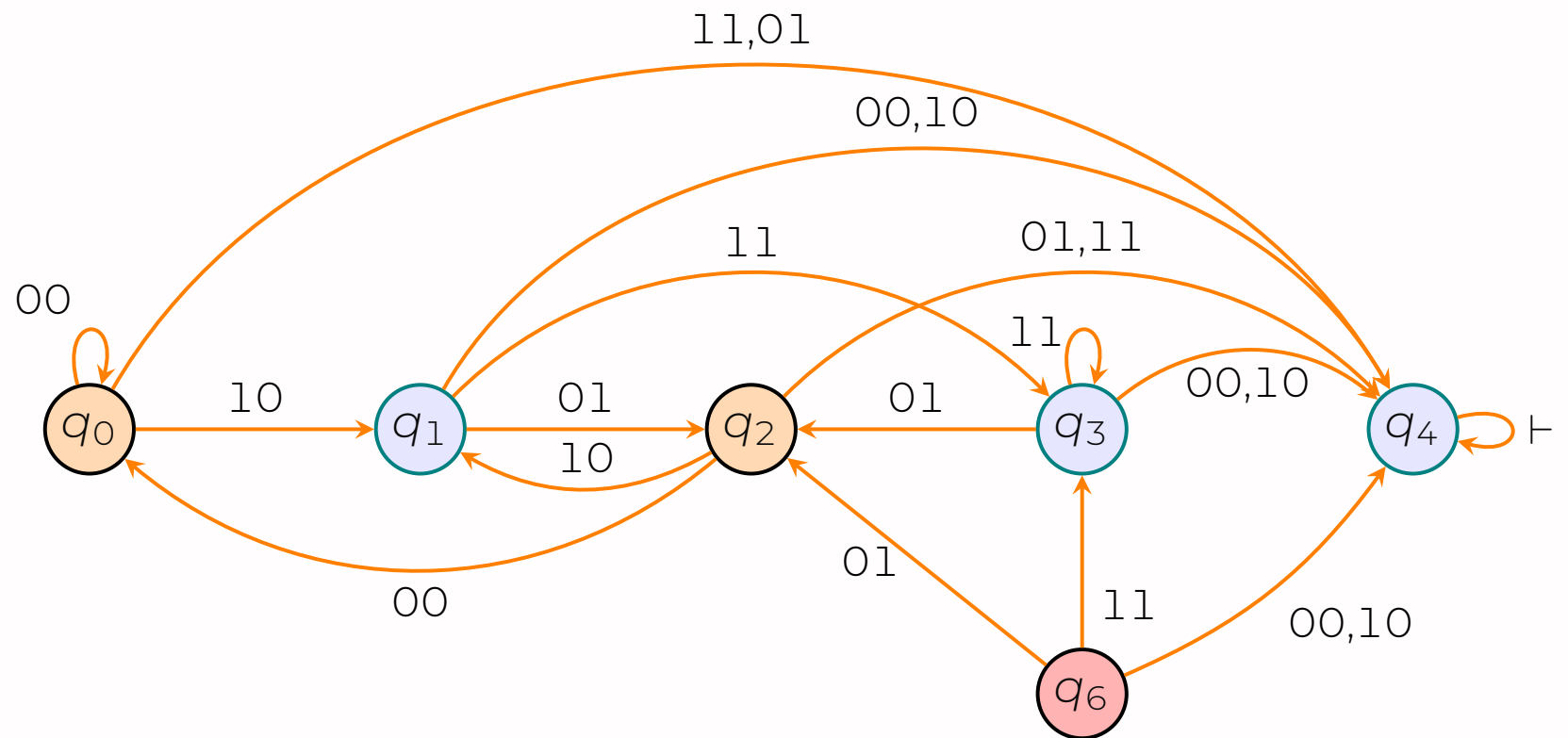
Each assignment corresponds to a row in a binary word.

Automata for Atomic Formulas

Construct finite automata over such binary words that accept all satisfying assignments for atomic formulas.

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Construction of automata for $y = 2x + 1$



accepting state $\rightarrow$ orange
 initial state $\rightarrow$ red
 What is Presburger?
 Presburger $\leftrightarrow$ Semilinear

> Why Presburger? > **Automata based Construction** >
 > CFL > Semilinear Sets > Decidability >

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Construction for Atomic Formulas

Consider the formula $\varphi: a_1x_1 + \dots + a_nx_n = b$, where $a_i \in \mathbb{Z}$:

Construct automaton $\mathcal{A}_\varphi$ as follows:

- Begin with the initial state labelled b .
- In general, if the current state is c , and the input is a bit vector $(\theta_1, \dots, \theta_n)$:
 - Check if $(a_1\theta_1 + \dots + a_n\theta_n) \equiv c \pmod{2}$.
 - If so, move to the state labelled

$$\frac{c - (a_1\theta_1 + \dots + a_n\theta_n)}{2}.$$

- Otherwise, move to the Junk state.
- Make the state labelled 0 the (only) final state.

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Claim on State Bound

The number of states is bounded by $2M + 2$ where

$$M = \max(|b|, |a_1| + \dots + |a_n|).$$

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Example 2

$$2x + y - 3z = 2$$

$$x \rightarrow 10001(17)$$

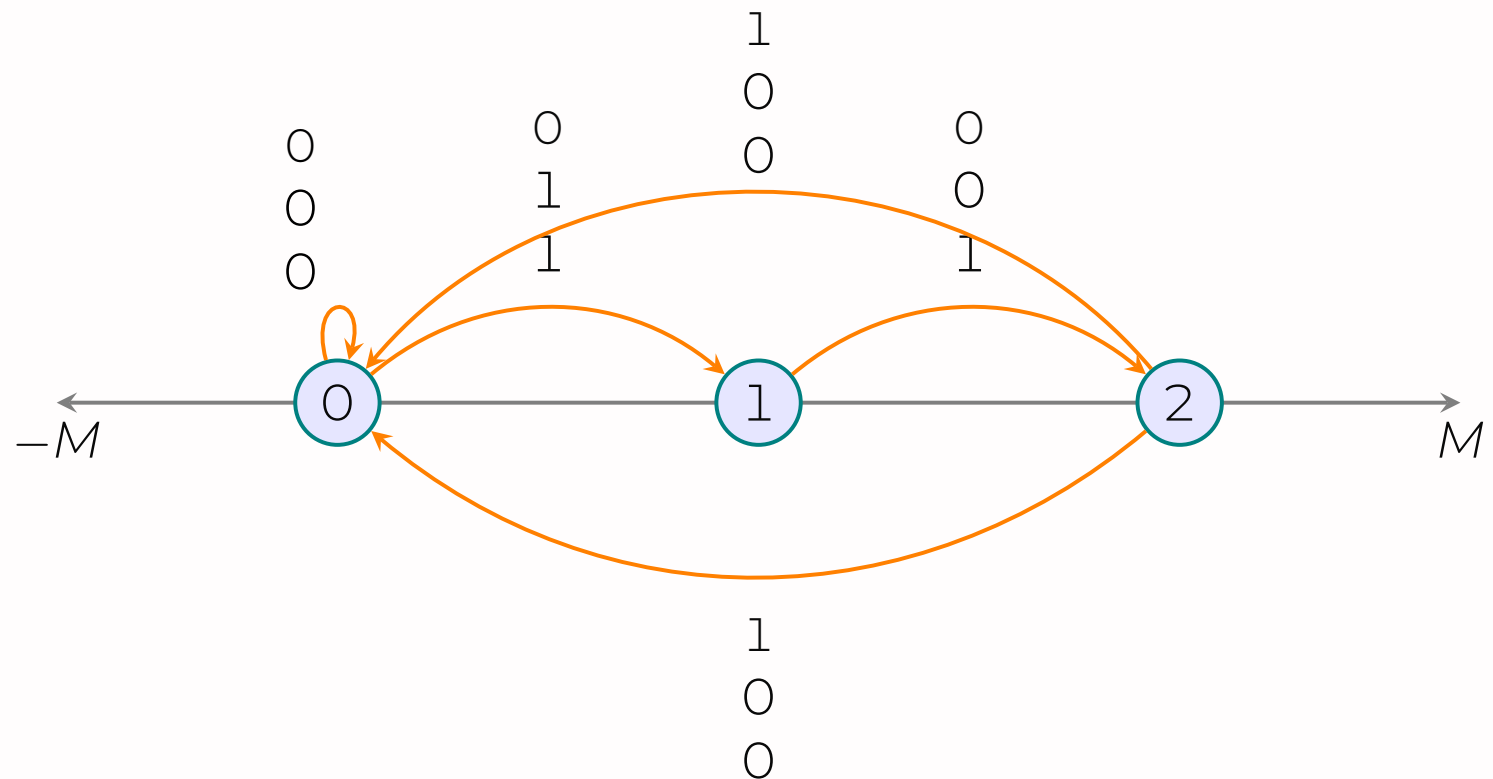
$$y \rightarrow 00100(04)$$

$$z \rightarrow 01100(12)$$

Keep track of the **weighted** sum **required** in the future to achieve the original weighted sum of b .

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Automata Transitions



◇ What is Presburger?
Presburger $\leftrightarrow$ Semilinear

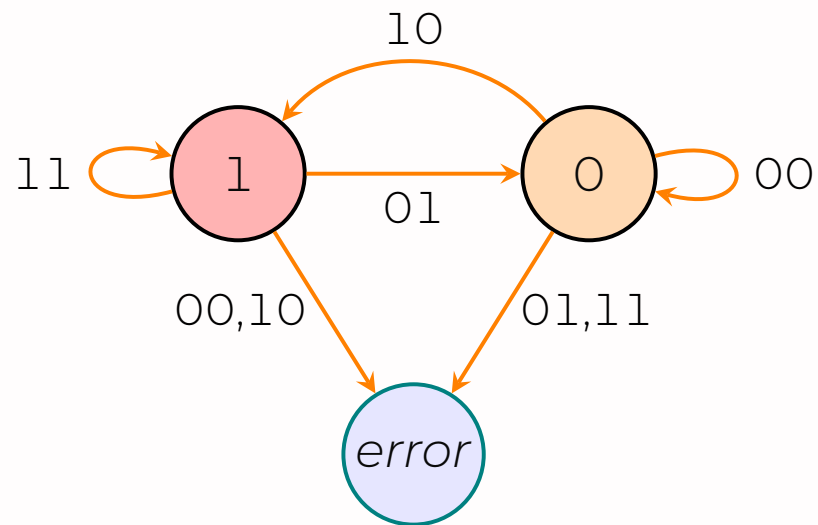
> Why Presburger?
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> **Automata based Construction**
> Semilinear Sets > Decidability

◇

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Construction of automata for $y = 2x + 1$



accepting state → orange
initial state → red

◇ What is Presburger?
Presburger ↔ Semilinear

> Why Presburger?
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> **Automata based Construction**
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◇

Key Insights

- An important class of subsets of lattice points in $\mathbb{N}^n$ arises from the theory of context-free languages — the **semilinear sets**.
- A set is **semilinear** if it is a finite union of cosets of finitely generated sub-semigroups of $\mathbb{N}^n$.
- These sets are shown to be **equivalent** to those definable by **modified Presburger formulas**.
- A **characterization** is provided for which semilinear sets correspond to languages (via Parikh images).
- Using this, a **decision procedure** is derived to determine whether a given linear set corresponds to a context-free language.

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Algebraic Structures in $\mathbb{N}^n$

- **Lattice Points:** Elements of $\mathbb{Z}^n$. We focus on the subset $\mathbb{N}^n$ (nonnegative coordinates).
- **Monoid:** A semigroup with an identity element. $\mathbb{N}^n$ under vector addition is a commutative monoid with identity 0.
- **Semigroup:** A set S with an associative binary operation. Here: $S \subseteq \mathbb{N}^n$, closed under addition.
- **Finitely Generated Semigroup:** Exists $v_1, \dots, v_k \in \mathbb{N}^n$ such that:

$$S = \left\{ \sum_{i=1}^k a_i v_i \mid a_i \in \mathbb{N} \right\}$$

Closed under addition by construction.

- **Subsemigroup:** $T \subseteq S$ that is also a semigroup (closed, associative).

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Semilinear Sets in $\mathbb{N}^n$

- **Coset (of Semigroup):** For $a \in \mathbb{N}^n$, and S a semigroup:

$$a + S = \{a + s \mid s \in S\}$$

- **Semilinear Set:** A finite union of cosets of finitely generated semigroups:

$$\bigcup_{i=1}^m (a_i + \langle v_{i1}, \dots, v_{ik_i} \rangle)$$

- **Properties:**
 - Closed under union, intersection, and complement (Boolean operations).
 - Every semilinear set is definable in Presburger arithmetic.
 - Intersection of two finitely generated semigroups is again finitely generated.

Internal Structure

- A **semilinear set** is a finite union of linear sets.
- Each linear set is defined by a **constant** $c \in \mathbb{N}^n$ and a finite set of **periods** $P \subseteq \mathbb{N}^n$.
- The set consists of all points of the form:

$$c + \lambda_1 p_1 + \cdots + \lambda_k p_k \quad (\lambda_i \in \mathbb{N})$$

- This is an **internal** description constructed using additive generators and base points.

Equivalence with Presburger Descriptions

Equivalence Theorem (Section 1)

- The class of semilinear sets is **identical** to the class of sets defined by modified Presburger formulas.
- Each internal (semilinear) description can be **effectively translated** into an external (Presburger) formula, and vice versa.
- More precisely, each set in the class is defined as the extension of a modified Presburger formula with n free variables, i.e., the set of all n -tuples of nonnegative integers satisfying the given formula.
- This mirrors a classical analogy in linear algebra:
 - **Internal**: A finite set of vectors that spans a subspace
 - **External**: A finite system of linear equations whose solution set defines the subspace

Semilinear Sets and Language Theory

- Semilinear sets arise naturally as **Parikh images** of formal languages.
- Our interest in semilinear sets stems from their relation to languages. Part III is devoted to this relation. It is shown that those semilinear sets which correspond to languages can be given semilinear descriptions of a particular form.
- For **linear sets**, a **decision procedure** is provided for determining whether an arbitrary set corresponds to a language.

Semigroup Structure

Let $\mathbb{N}$ be the set of nonnegative integers, and let $\mathbb{N}^n$ be the Cartesian product of $\mathbb{N}$ with itself n times.

For $x = (x_1, \dots, x_n), y = (y_1, \dots, y_n)$ in $\mathbb{N}^n$:

$$x + y = (x_1 + y_1, \dots, x_n + y_n)$$

For $t \in \mathbb{N}$:

$$t \cdot x = (tx_1, \dots, tx_n)$$

Then $\mathbb{N}^n$ forms a commutative semigroup (under addition), and a partial order is defined by:

$$x \leq y \iff x_i \leq y_i \text{ for all } 1 \leq i \leq n$$

Linear Sets

Given subsets $C, P \subseteq \mathbb{N}^n$, define:

$$L(C; P) = \{x_0 + x_1 + \cdots + x_m \mid x_0 \in C, x_i \in P\}$$

This describes all elements generated by adding a constant x_0 and a finite sequence of periods from P .

If $C = \{c\}$ and $P = \{p_1, \dots, p_r\}$, we write:

$$L(c; P) = L(\{c\}; P)$$

A subset $L \subseteq \mathbb{N}^n$ is called **linear** if $L = L(c; P)$ for some $c \in \mathbb{N}^n$, $P \subseteq \mathbb{N}^n$.

Linear and Semilinear Sets

Let $P \subseteq \mathbb{N}^n$. The set P generates a finitely generated sub-semigroup $S \subseteq \mathbb{N}^n$, and for any $c \in \mathbb{N}^n$, the set:

$$L(c; P) = \left\{ c + \sum_{i=1}^k a_i p_i \mid a_i \in \mathbb{N}, p_i \in P \right\}$$

is a **coset** of S in $\mathbb{N}^n$ containing c .

Thus, a set $L \subseteq \mathbb{N}^n$ is **linear** if and only if it is a coset of a finitely generated sub-semigroup of $\mathbb{N}^n$.

Semilinear Sets

A subset of $\mathbb{N}^n$ is called **semilinear** if it is a finite union of linear sets.

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A Non-Semilinear Sub-semigroup in $\mathbb{N}^2$

Definition

Let:

$$X = \{(x, y) \in \mathbb{N}^2 \mid y \leq x^2\}$$

31.2

A Non-Semilinear Sub-semigroup in $\mathbb{N}^2$

Definition

Let:

$$X = \{(x, y) \in \mathbb{N}^2 \mid y \leq x^2\}$$

Every vertical line $x = k$ intersects X in a finite set:

$$X \cap \{k\} \times \mathbb{N} = \{(k, y) \mid y \leq k^2\}$$

Thus, X cannot contain any linear set with a period of the form $(0, y)$.

Why X is Not Semilinear (1)

Assumption

Suppose:

$$X = \bigcup_{i=1}^m L(c_i; P_i)$$

where each $L(c_i; P_i)$ is a linear set.

Since each vertical line intersects X in a finite set, no period $p \in P_i$ can be of the form $(0, y)$.

Hence, all periods must satisfy $x > 0$. Let:

$$M = \max \left\{ \frac{y}{x} \mid (x, y) \in \bigcup P_i, x > 0 \right\}$$

Why X is Not Semilinear (2)

Contradiction via Slope

Pick two points in X : (z_1, z_1^2) , (z_2, z_2^2) with $z_1, z_2 > M$, where

$$M = \max \left\{ \frac{y}{x} \mid (x, y) \in \bigcup P_i, x > 0 \right\}$$

Then the slope between them is:

$$\frac{z_2^2 - z_1^2}{z_2 - z_1} = z_2 + z_1 > 2M$$

which exceeds all period slopes.

Lemma 6.2

- If C_1, C_2 , and P are subsets of N^n , then $L(C_1 \cup C_2; P) = L(C_1; P) \cup L(C_2; P)$.
- If C, P_1 , and P_2 are subsets of N^n , then $L(L(C; P_1); P_2) = L(C; P_1 \cup P_2)$.
- If C_1, P_1 are subsets of N^n and C_2, P_2 subsets of N^m , then $L(C_1; P_1) \times L(C_2; P_2) = L(C_1 \times C_2; (P_1; 0^m) \cup (0^n \times P_2))$

Lemma 6.3

Let t be a linear function of N^n into N^m . If X is a linear subset of N^n , then $\tau(X)$ is a linear subset of N^m . If X is a semi-linear subset of N^n then $\tau(X)$ is a semi-linear subset of N^m

Lemma 6.4

It is solvable to determine for arbitrary $u_i, (1 \leq i \leq p)$ and $v_j (1 \leq j \leq q)$ be in N^n , and let w be a arbitrary n-tuple w of integers whether there exists a positive solution $w = \sum_{i=1}^p a_i u_i - \sum_{j=1}^q b_j v_j$.

Lemma 6.1

- Each set of pairwise incomparable elements of N^n is finite.
- Each subset of N^n has only a finite number of minimal elements.

Lemma 6.5

Let $u_i, (1 \leq i \leq p)$ and $v_j (1 \leq j \leq q)$ be in N^n , and let w be a fixed n -tuple of integers. Then it is solvable to determine all minimal positive solutions to the equation

$$w = \sum_{i=1}^p a_i u_i - \sum_{j=1}^q b_j v_j$$

Theorem 6.1

Let X and X' be semi-linear subsets of N^n . Then $X \cap X'$ is a semi-linear subset of N^n and is effectively calculable from X and X' .

Theorem 6.2

If X and Y are semi-linear subsets of N^n , then $X - Y$ is also a semi-linear subset of N^n and is effectively calculable from X and Y .

Theorem 1

The family of semilinear sets of $\mathbb{N}^n$ is closed with respect to union, intersection, and complementation. The projection of a semilinear set is semilinear.

Presburger Formulas

We consider formulas about nonnegative integers. If P is such a formula and has $n \geq 1$ free variables $x_1, \dots, x_n$, we also write it as $P(x_1, \dots, x_n)$.

The set of Presburger formulas, denoted by $\mathcal{P}$, is the smallest class of formulas satisfying the following five conditions:

Presburger Formulas - Conditions

1. For given nonnegative integers $t_i, t'_i, 0 \leq i \leq n$,

$$t_0 + \sum_{i=1}^n t_i x_i = t'_0 + \sum_{i=1}^n t'_i x_i$$

is a formula in $\mathcal{P}$.

2. If P_1, P_2 are in $\mathcal{P}$, so is their conjunction $P_1 \wedge P_2$.
3. If P_1, P_2 are in $\mathcal{P}$, so is their disjunction $P_1 \vee P_2$.

- 4 If P is in $\mathcal{P}$, so is its negation $\sim P$.
- 5 If $P(x_1, \dots, x_n)$ is in $\mathcal{P}$ and $1 \leq i \leq n$, then the formula $(\exists x_i)P(x_1, \dots, x_n)$ is in $\mathcal{P}$.

A Presburger sentence is defined to be a Presburger formula with no free variables.

Decidability of Presburger Arithmetic

Theorem 2

It is decidable whether an arbitrary Presburger sentence is true.

Remark 1

$$t_0 + \sum_{i=1}^n t_i x_i \leq t'_0 + \sum_{i=1}^n t'_i x_i$$

is equivalent to the Presburger formula:

$$(\exists x_{n+1} \in \mathbb{N}) \left(t_0 + \sum_{i=1}^n t_i x_i + x_{n+1} = t'_0 + \sum_{i=1}^n t'_i x_i \right)$$

More Presburger Formula Equivalences

Remark 2

If $P(x_1, \dots, x_n)$ is a Presburger formula, then the formula

$$(\forall x_i)P(x_1, \dots, x_n)$$

is regarded as a Presburger formula since it is equivalent to

$$\sim (\exists x_i)(\sim P(x_1, \dots, x_n)).$$

Similarly if P and Q are Presburger formulas, $P \Rightarrow Q$ is regarded as a Presburger formula.

Presburger Sets

For $n > 0$, a set A is a Presburger set in $\mathbb{N}^n$ if

$$A = \{(x_1, \dots, x_n) \mid P(x_1, \dots, x_n) \text{ is true}\}$$

for some $P(x_1, \dots, x_n)$ in $\mathcal{P}$. A Presburger set is a Presburger set in $\mathbb{N}^n$ for some n .

It follows from (2), (3), and (4) that the family of Presburger sets in $\mathbb{N}^n$ is closed under intersection, union, and complementation. From (5), the projection of a Presburger set is a Presburger set.

Equivalence of Presburger and Semilinear Sets

Theorem 3

The family of Presburger sets of $\mathbb{N}^n$ is identical with the family of semilinear sets of $\mathbb{N}^n$. Furthermore, each description is effectively calculable from the other.

Semilinear Sets are Presburger Sets

Sketch of the Proof

- A linear set $L = L(v_0; \{v_1, \dots, v_k\}) \subseteq \mathbb{N}^n$ can be described by a Presburger formula.
- Introduce existential variables $a_1, \dots, a_k \in \mathbb{N}$ and express:

$$x_j = v_{0j} + \sum_{i=1}^k a_i v_{ij}, \quad \text{for each } j = 1, \dots, n$$

- The formula uses only addition, equality, and quantifiers — all allowed in Presburger arithmetic.
- A semilinear set is a finite union of such linear sets, so use disjunction to combine their formulas.

Example: Linear Set in $\mathbb{N}^3$

Definition

Let $L = L(c; \{p_1, p_2\}) \subseteq \mathbb{N}^3$, where:

- $c = (1, 0, 2)$ is the constant vector
- $p_1 = (1, 2, 0)$, $p_2 = (0, 1, 1)$ are the period vectors

Then L consists of all points:

$$x = c + a_1 p_1 + a_2 p_2, \quad \text{with } a_1, a_2 \in \mathbb{N}$$

i.e.,

$$x_1 = 1 + a_1, \quad x_2 = 2a_1 + a_2, \quad x_3 = 2 + a_2$$

Presburger Formula

$$P_L(x_1, x_2, x_3) := (\exists a_1)(\exists a_2) [x_1 = 1 + a_1 \wedge x_2 = 2a_1 + a_2 \wedge x_3 = 2 + a_2]$$

Presburger Sets are Semilinear Sets

Sketch of Proof

By Theorem 1.1, it suffices to show that the set of nonnegative integer solutions of the equation:

$$t_0 + \sum_{i=1}^n t_i x_i = t'_0 + \sum_{i=1}^n t'_i x_i \quad (1)$$

is semilinear.

Minimal Solutions

Let:

- $C \subseteq \mathbb{N}^n$: the set of minimal solutions of equation (1)
- $P \subseteq \mathbb{N}^n \setminus \{0^n\}$: the set of minimal solutions of the associated homogeneous equation:

$$\sum_{i=1}^n t_i x_i = \sum_{i=1}^n t'_i x_i \quad (2)$$

Both sets C and P are finite and effectively computable.

Presburger Sets are Semilinear

Construction

By Corollary of Lemma 6.1 (from [1]), both C and P are finite. By Lemma 6.5, they are also effectively computable.

It follows from Theorem 6.1 that the set of nonnegative solutions to (1) is:

$$\bigcup_{c \in C} L(c; P)$$

where each linear set $L(c; P)$ is defined by:

$$L(c; P) = \{c + a_1 p_1 + \cdots + a_k p_k \mid a_i \in \mathbb{N}, p_i \in P\}$$

Thank You!

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